Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS at USC with 2+ years building **C++-adjacent systems**, **Python tooling**, and low-latency data workflows, with experience in **Linux**, simulation, concurrency, and production reliability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **Python-first platform services** for 500+ users, reducing request latency from **1.2s to 350ms** by optimizing execution paths, tuning SQL access patterns, and improving responsiveness for analytics workflows.
- Implemented **automated validation and ETL checks** for 5,000+ monthly jobs, reducing downstream failures while improving runtime observability, release confidence, and reliability across recurring production data-processing workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **backend features and APIs** handling 10,000+ daily requests, translating product requirements into maintainable services while improving delivery predictability through clearer implementation ownership, sequencing, and fault-aware execution.
- Built **AI-assisted grading workflows** with multimodal models and OCR, reducing feedback turnaround by **60%** while supporting production operations with concurrent batch execution, cost tracking, and CI/CD-backed releases.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **large-scale simulation pipelines** across 100,000+ trajectory records, enabling reproducible algorithm evaluation, faster diagnostics, and high-confidence benchmarking while improving signal quality for debugging model-driven decision workflows.
- Architected a **spatial-temporal encoder** integrating HD-map geometry with agent context, improving closed-loop planning stability by **15%** while strengthening evaluation rigor for long-horizon autonomous-system experiments.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **API and ETL microservices** using AWS Glue and S3, improving response time by **30%** while supporting distributed integrations and cross-platform validation workflows across Windows, Linux, and macOS environments.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and improving confidence in frequent releases across coordinated service components.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Built **Flask booking and payment APIs** for a cross-platform services product, supporting 50K+ booking records while maintaining sub-200ms core flows, real-time synchronization, and consistent transaction-state updates.
- Integrated **payment and refund webhooks** with automation, reducing manual handling by **70%** while improving transaction visibility, coupon validation, and operational reliability across customer-facing booking workflows.

Skills

Languages: **C++**, **Python**, Java, JavaScript/TypeScript, SQL, Bash
Systems: Linux/Unix, TCP/IP, concurrency fundamentals, distributed systems, performance optimization
Backend & APIs: Flask, FastAPI, Node.js, REST APIs, ETL pipelines, async workflows
Databases: PostgreSQL, MySQL, MongoDB, SQL Server, Firebase
Cloud & Infrastructure: AWS, Azure, Docker, Kubernetes, Jenkins, CI/CD
Engineering: automated testing, debugging, technical documentation, Agile/Scrum, cross-functional collaboration

Projects

CPU-scheduling Simulator Live | *C++, React.js, JavaScript*

- Implemented **FCFS, SJF, SRT, RR, priority, and MLQ** algorithms in a **C++**-backed simulator, pairing React controls with Gantt visualization across 1K+ process combinations.

FinNexus Live | *Next.js, TypeScript, MongoDB, Plaid*

- Built a **fintech platform** supporting secure bank linking, real-time transaction visibility, and advanced filtering across 1M+ transactions for 50,000+ users through scalable TypeScript-backed workflows.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Built a **containerized REST processing API** on AWS CodePipeline and ECS/Fargate with automated validation gates, improving reliability, reducing deployment variance, and streamlining troubleshooting across production workflows.